

Freight Rate Prediction Assessment

Leakage-safe model development and submission artifacts

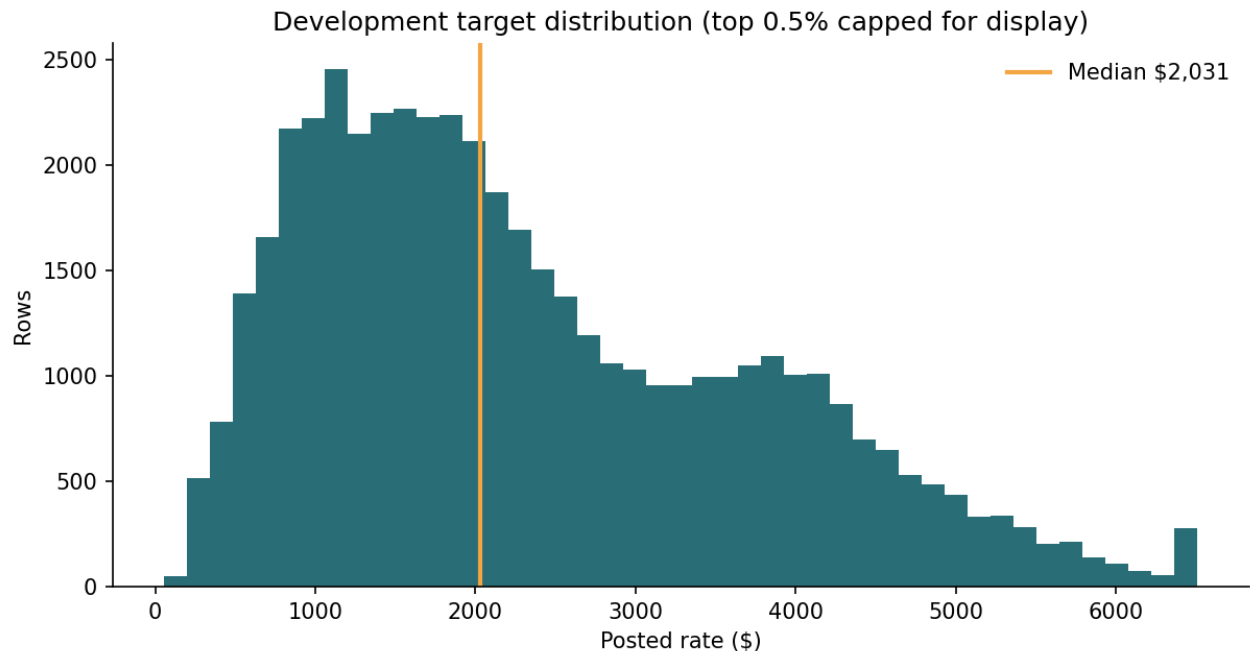
Executive summary. HistGradientBoostingRegressor candidate **hgb_compact** using `basic_plus_geographic` achieved a blocked future-holdout MAE of \$129.51, RMSE \$640.32, R-squared 0.8239, and sMAPE 4.527%. The model was refit on all 48,000 labeled rows to generate the 12,000-row submission and was saved as `models\final_rate_model.joblib`. No external data or final-validation labels were used.

MAIN|model=hgb_compact|family=HistGradientBoostingRegressor|feature_set=basic_plus_geographic|mae=129.511954241217

1. Dataset and data quality

Development contains 48,000 rows from 2025-01-01 to 2025-10-31; final validation contains 12,000 later rows from 2025-11-01 to 2025-12-31. The target is `posted_rate` (median \$2,030.76, mean \$2,373.98, maximum \$25,533.00). There are 64 development pickup cities, 64 delivery cities, and three equipment types.

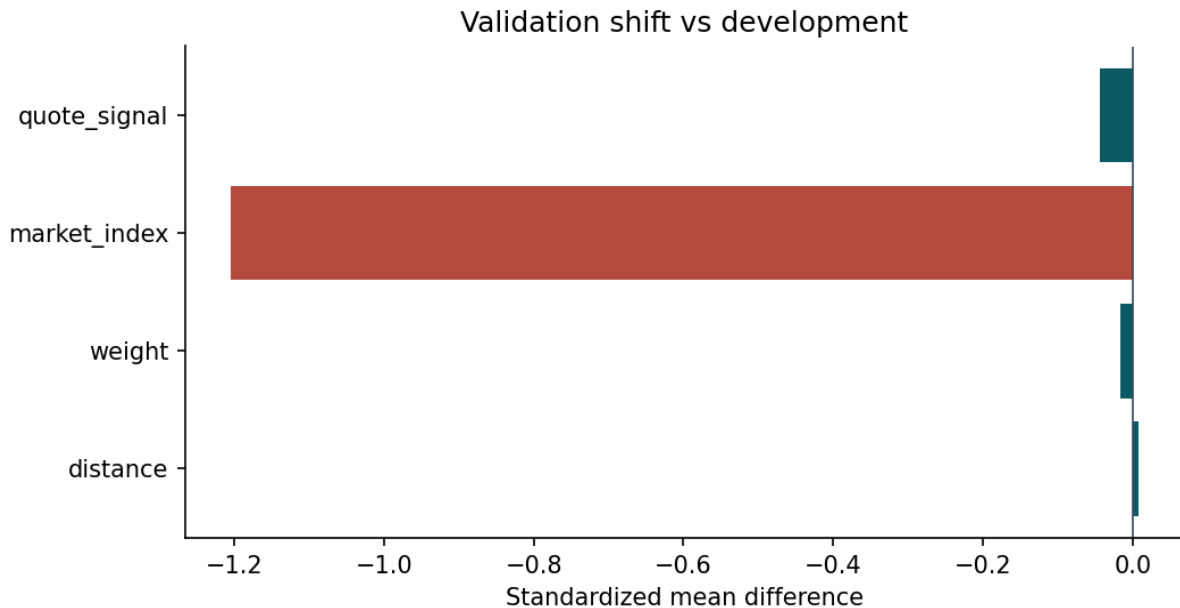
No duplicate IDs, exact duplicate rows, invalid dates, or illegal coordinates were found. Development has 300 missing weights, 374 missing market-index values, and 292 negative weights. The pipeline treats impossible non-positive weights as missing and retains explicit missing and problem flags. Learned preprocessing is fitted only on the applicable training rows. Valid rate extremes are retained and the MAE loss reduces their influence without hiding them.



2. Validation design and exploratory findings

The final data is strictly later than development, so the primary split trains on 2025-01-01 through 2025-08-31 (38,477 rows) and validates on 2025-09-01 through 2025-10-31 (9,523 rows). This 61-day blocked holdout mirrors the final November-December horizon. A separate training-only chronological window fits through 2025-07-01 and validates from 2025-07-02 through 2025-08-31 for feature screening. Preprocessing statistics and frequency maps use training rows only.

Distance is the dominant supplied signal. Market index has the clearest development-to-validation drift; the September-October regime is much closer to final validation than the full historical average. Final validation contains eight unseen cities and 1,461 unseen-route rows, a limitation not fully represented by the holdout because all holdout cities were already observed.



3. Cleaning and features

Categories are whitespace/case normalised and missing categories become Unknown. Numeric missing values remain missing and are accompanied by explicit flags; learned preprocessing stays within training rows. Tested groups cover raw supplied fields, cyclic and calendar date features, route combinations with training-only frequencies/unseen flags, compact geographic calculations, and defensible distance/weight/market interactions. load_id is never a model feature. Coordinates appear synthetic, so supplied distance remains authoritative.

4. Model comparison and selection

MAE is the primary selection metric because dollar error is directly actionable and the labels contain sparse extremes. Fixed HGB and CatBoost proxies scored every feature group on the same training-internal chronological window. The lowest equal-family mean MAE froze one logical feature list for every eligible finalist, and CatBoost's best iteration came from that fold-pure screen. Every eligible HGB and CatBoost candidate then used the same outer chronological rows, frozen feature columns, rate-per-mile target, and posted-rate MAE. Every configuration was frozen before outer scoring; outer labels made the required final candidate choice and supported diagnostics, without adapting a candidate before its MAE was recorded. The documented rule was: Lowest finite MAE on the common chronological holdout among all eligible candidates; exact-equality ties are broken deterministically by candidate name. RMSE, R-squared, and sMAPE are reported for context.

Model	Family	Features	MAE	RMSE	Eligible	Selected
HGB compact	HGB	Geographic	129.51	640.32	Yes	Yes
HGB smoother	HGB	Geographic	130.26	640.40	Yes	No
CatBoost	CatBoost	Geographic	141.07	643.10	Yes	No
Equipment baseline	Baseline	Distance + equipment	229.10	670.66	No	No
Global baseline	Baseline	Target only	1148.92	1569.42	No	No

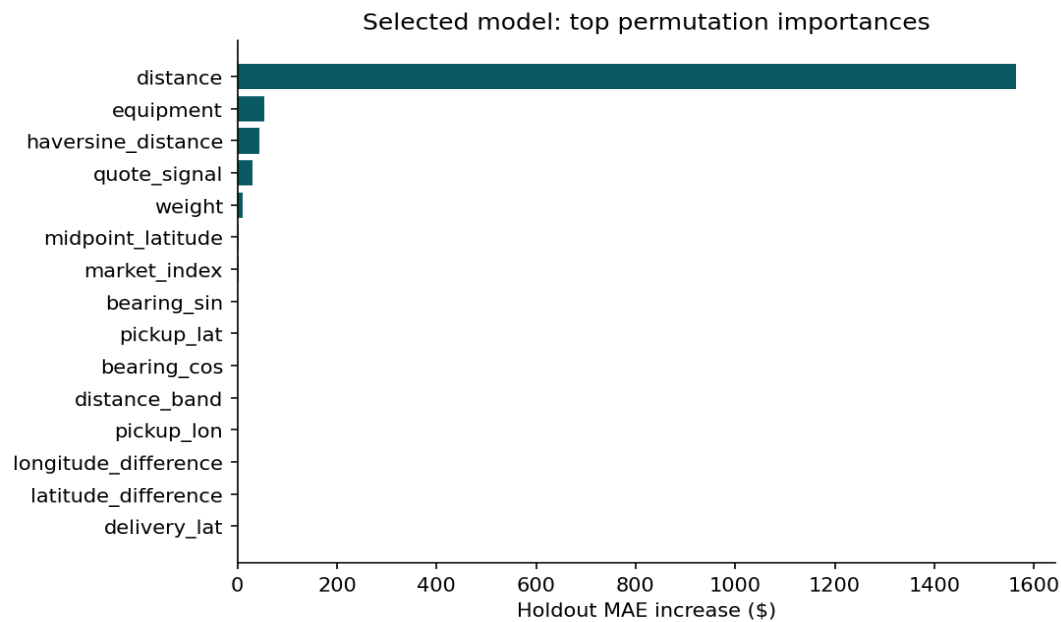
Training-only family-neutral feature screening (positive change improves on basic)

Feature set	HGB MAE	CatBoost MAE	Mean MAE	Change	Selected
basic_plus_geographic	134.89	119.71	127.30	6.36	Yes
basic_plus_route	134.82	124.56	129.69	3.97	No
basic_supplied	139.34	127.98	133.66	0.00	No
basic_plus_interactions	139.03	129.26	134.15	-0.49	No
best_combined	134.63	149.07	141.85	-8.19	No
basic_plus_date	137.56	153.46	145.51	-11.85	No
compact_calendar	167.76	147.85	157.80	-24.15	No

Selected: **HistGradientBoostingRegressor** candidate **hgb_compact** with **basic_plus_geographic**. Histogram gradient boosting uses an ordinal category encoder fitted within each training fit; previously unseen categories map to a reserved value. The leading importance features were distance, equipment, haversine_distance, quote_signal, weight. Parameters: {"l2_regularization": 3.0, "learning_rate": 0.06, "max_iter": 250, "max_leaf_nodes": 15, "min_samples_leaf": 30}. Saved model: models\final_rate_model.joblib.

5. Error analysis and model interpretation

The table shows the highest-error reported segments. Rare extreme labels drive RMSE and high-target error. Route/city segmentation is retained in the artifacts for inspection; final unseen-city error may be higher than this holdout because the new cities arrive only in the unlabeled future set.



Dimension	Segment	Rows	MAE	Bias
target_band	High	2,381	311.40	-305.27
distance_band	Very Long	2,460	232.08	-190.16
distance_band	Long	3,157	147.09	-123.13
equipment	Reefer	2,393	145.92	-117.54
route_status	Common (>=10)	5,900	134.57	-107.72
equipment	Flatbed	1,770	131.71	-98.24

6. Reduced-schema December predictions

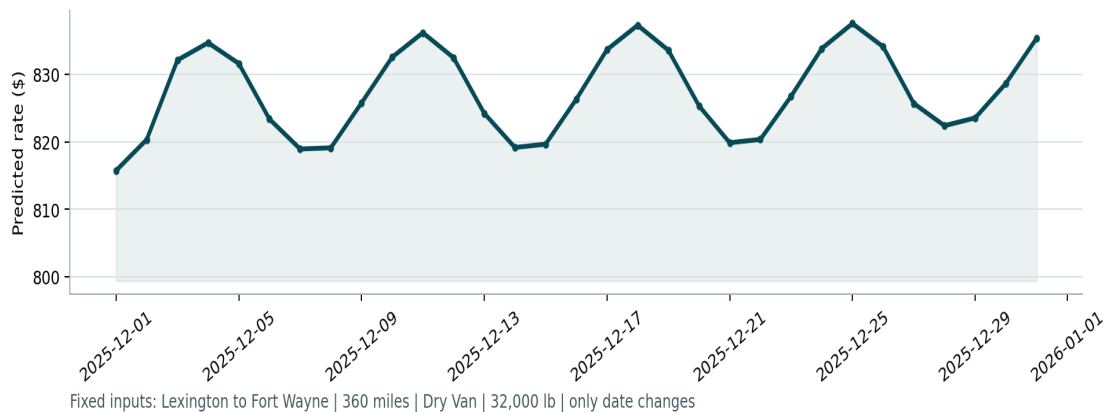
This selection is separate from the main validation model. Every eligible reduced-schema HGB and CatBoost candidate used the same outer chronological rows, rate-per-mile target, and posted-rate MAE; within each feature-set alternative, both families received identical compatible columns. Any CatBoost early stopping used only an internal chronological split within the outer training rows.

Model	Family	Features	MAE	RMSE	Eligible	Selected
HGB basic	HGB	Basic	108.78	634.41	Yes	Yes
HGB geographic	HGB	Geographic	110.24	634.85	Yes	No
CatBoost geographic	CatBoost	Geographic	125.70	634.78	Yes	No
CatBoost basic	CatBoost	Basic	126.04	634.98	Yes	No

Selected: **HistGradientBoostingRegressor** candidate **december_hgb_december_basic** using **december_basic**, holdout MAE **\$108.78**. It uses only pickup, delivery, distance, equipment, weight, known calendar fields and, if selected, development-derived city coordinates. Market and quote signals are not fabricated. A separate model is refit on all labeled rows. Parameters: {"l2_regularization": 3.0, "learning_rate": 0.06, "max_iter": 250, "max_leaf_nodes": 15, "min_samples_leaf": 30}. Saved model: models\december_rate_model.joblib.

DECEMBER|model=december_hgb_december_basic|family=HistGradientBoostingRegressor|feature_set=december_basic|mae=108.7836047385515

Candidate: December 2025 Predicted Load Rate



7. Limitations and reproduction

Future market regimes, unseen cities/routes, synthetic coordinates, sparse abnormal labels, and reliance on two proxy families plus one training-only common-feature window remain limitations. The outer-holdout MAE is a model-selection estimate, not a post-selection unbiased test score. Install with `python -m pip install -r requirements.txt` and reproduce everything with `python run_pipeline.py`. The unmodified scorer validates both outputs and creates the chart above; it does not calculate accuracy.